

$$\begin{aligned} \text{For } (A < B), \quad & X = \bar{A}_0 B_0 \\ \text{For } (A = B), \quad & Y = \bar{A}_0 \bar{B}_0 + A_0 B_0 = \overline{A_0 \oplus B_0} \\ \text{For } (A > B), \quad & Z = A_0 \bar{B}_0 \end{aligned}$$

Logic Diagram of 1-bit Comparator:

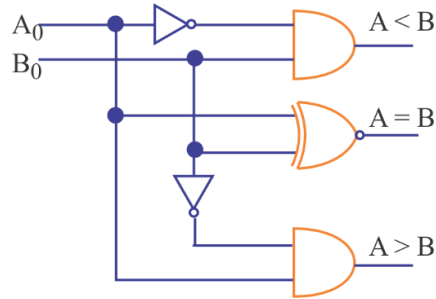


Fig. 3.17. Logic Diagram of 1-bit Comparator

Example: The circuit shown in given figure is:

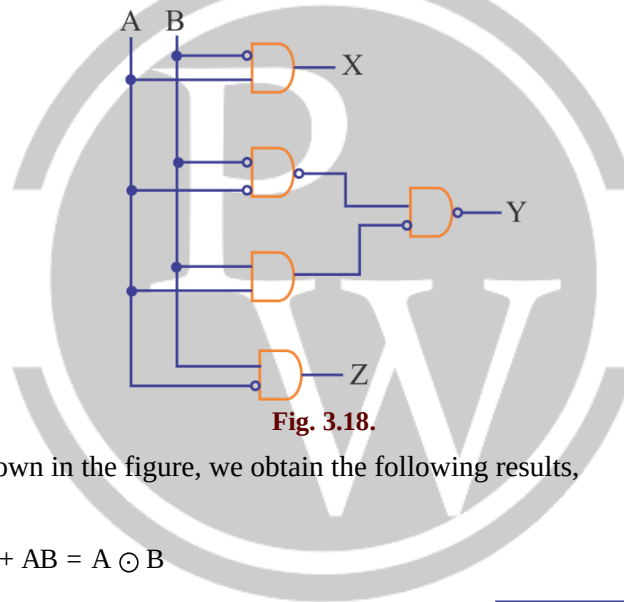


Fig. 3.18.

Solution: From the logic circuit shown in the figure, we obtain the following results,

$$X = A\bar{B}$$

$$Y = \overline{(A\bar{B})(AB)} = \bar{A}\bar{B} + AB = A \odot B$$

$$Z = \bar{A}B$$

So, we obtain the truth table for the above function as shown below.

From truth table, we deduce the following results.

It $A > B$, then $x = 1$

It $A = B$, the $y = 1$

It $A < B$, then $z = 1$

Therefore, it is a comparator circuit.

A	B	X	Y	Z
0	0	0	1	0
0	1	0	0	1
1	0	1	0	0
1	1	0	1	0

3.7. Multiplexer

A multiplexer, abbreviated as MUX, is a digital switch which selects one of the many inputs to a single output. A number of control lines determine which input data is to be routed to the output. If there are n select lines, then the number of maximum possible input lines is 2^n and the multiplexer is referred to as a 2^n -to-1 multiplexer or $2^n \times 1$ multiplexer.

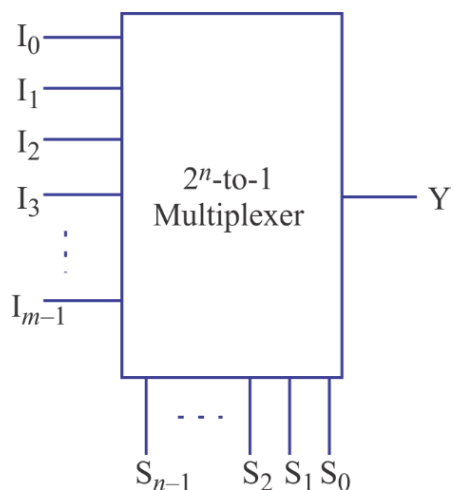


Fig. 3.19. Block diagram of a 2^n to 1 multiplexer

3.7.1. 2×1 MUX

A 2 to 1 multiplexer has 2 inputs. Since $2 = 2^1$, this multiplexer will have one control (select) line. It has two data inputs I_0 and I_1 , one select input S , and one output.

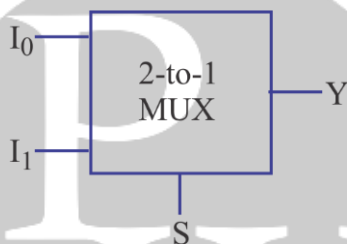


Fig. 3.20. Schematic block diagram of 2:1 Multiplexer

The truth table of this MUX is given below,

Select Line (S)	Output Y
0	I_0
1	I_1

Thus, the SOP expression for the output Y is,

$$Y = I_0 \bar{S}_0 + I_1 S_0$$

Realization of a 2:1 MUX using Logic Gates:

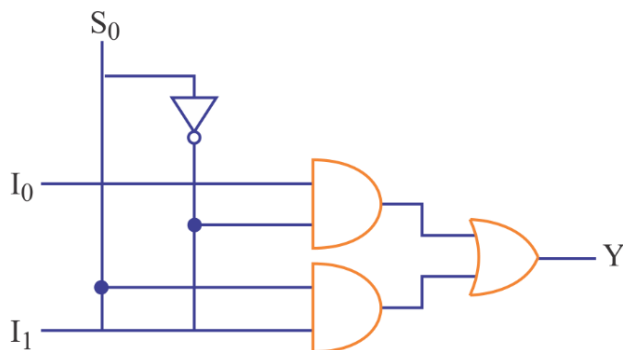


Fig. 3.21. Logic Diagram of a 2×1 Multiplexer

2.7.2. 4 × 1 MUX

A 4-to-1 multiplexer has 4 inputs and two select lines, where I_0 to I_3 are the four inputs to the multiplexer, and S_0 and S_1 are the select lines.

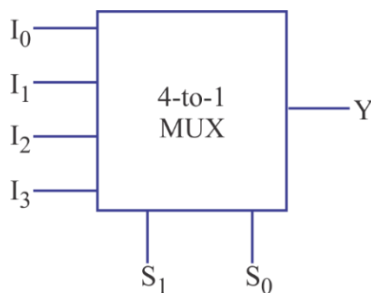


Fig. 3.22. Schematic block diagram of 4 × 1 MUX

Truth Table of a 4-to-1 Multiplexer

Select Inputs		Output
S_1	S_0	Y
0	0	I_0
0	1	I_1
1	0	I_2
1	1	I_3

Output Y for a 4-input multiplexer is

$$Y = I_0 \bar{S}_1 \bar{S}_0 + I_1 \bar{S}_1 S_0 + I_2 S_1 \bar{S}_0 + I_3 S_1 S_0$$

3.8. Implementation of Higher Order Mux Using Lower Order MUX

The methods for implementing higher order MUX using lower order MUX are

- Step 1:** If 2^n is the number of input lines in the available lower order multiplexer and 2^N is the number of input lines in the desired multiplexer, then the number of lower order multiplexers required to construct the desired multiplexer circuit would be $2^N - n$.
- Step 2:** From the knowledge of the number of selection inputs of the available multiplexer and that of the desired multiplexer, connect the less significant bits of the selection inputs of the desired multiplexer to the selection inputs of the available multiplexer.
- Step 3:** The most significant bits of the selection inputs of the desired multiplexer circuit are used to enable or disable the individual multiplexers so that their outputs when OR produce the final output.

Example: In realization of 32 : 1 MUX using 2 : 1 MUX, the required number of 2 : 1 MUX is ?

Solution: In realization of 2^n : 1 MUX using 2 : 1 MUX, the required number of 2 : 1 MUX is $2^n - 1$, since, we have to realize 32 : 1 MUX, so we have

$$n = 5$$

Hence, the required number of 2 : 1 MUX is

$$2^n - 1 = 2^5 - 1 = 31$$

3.9. Applications of Multiplexers

1. It is used as a data selector to select one out of many data inputs.
2. They are used in designing the combinational circuits.
3. They are used in digital-to-analog and analog-to-digital converters.
4. They can be used for simplification of logic design.
5. Multiplexers are also used in data acquisition systems.

3.10. Demultiplexer

The demultiplexer is a combinational logic circuit that performs the reverse operation of a multiplexer. The demultiplexer has one input line and m output lines. Again $m = 2^n$, so it requires n select lines. A demultiplexer with one input and m output is called a 1-to- m demultiplexer.

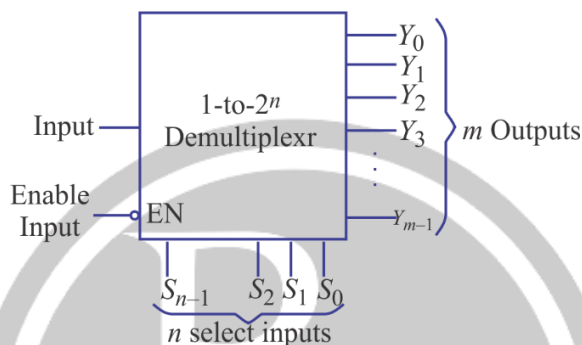


Fig. 3.23. Block Diagram of m-output Demultiplexer

The demultiplexer has one input line and m output lines. Again $m = 2^n$, so it requires n select lines. A demultiplexer with one input and m outputs is called a 1-to- m demultiplexer.

3.10.1. 1×2 Demultiplexer

A 1 to 2 demultiplexer has one input and two outputs. Since $2 = 2 \times 1$, it requires only one control (select) line.

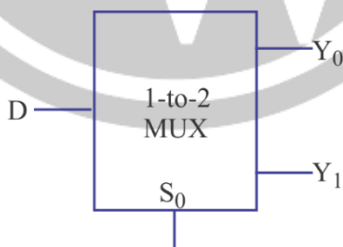


Fig. 3.24. Logic Diagram of 1×2 De-MUX

Truth table of a 1-to-2 demultiplexer

Table: Truth table of a 1-to-2 demultiplexer

Input S	Select input S ₀	Output	
		Y ₁	Y ₀
D	0	0	D
D	1	D	0

Thus, the Boolean expressions for the outputs can be written as

$$Y_0 = D\bar{S}_0 \text{ \& \; } Y_1 = DS_0$$

Realization of a 1 x 2 Demultiplexer using Logic Gates:

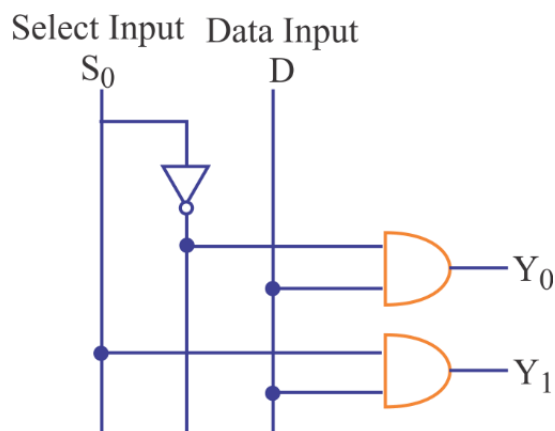


Fig. 3.25. Logic Diagram of 1 × 2 Demultiplexer

3.10.2. Applications of Demultiplexers

Demultiplexers are used in

1. Data transmission
2. Implementation of Boolean Functions
3. Combinational logic circuit design
4. Generate enable signals (enable one out of many). The application of enable signals in microprocessor systems are:
 - (a) Selecting different banks of memory
 - (b) Selecting different input/output devices for data transfer
 - (c) Enabling different functional units
 - (d) Enabling different rows of memory chips depending on address

3.11. Comparison Between Multiplexer and Demultiplexer

Table : Comparison between Multiplexer and Demultiplexer

S.No.	Parameter of comparison	Multiplexer	Demultiplexer
1.	Type of logic circuit	Combinational	Combinational
2.	Number of data inputs	m	1
3.	Number of select inputs	n	N
4.	Number of data output	1	M
5.	Relation between input/output lines and select lines	$m = 2^n$	$M = 2^N$
6.	Operation principle	Many to 1 or as data selector	1 to many or data distributor

3.12. Decoder

A decoder is a combinational circuit that converts an n-bit binary input data into 2^n output lines, such that each output line will be activated for only one of the possible combinations of inputs. Decoders are usually represented as n-to- 2^n line decoders, where n is the number of input lines and 2^n is the number of maximum possible output lines.

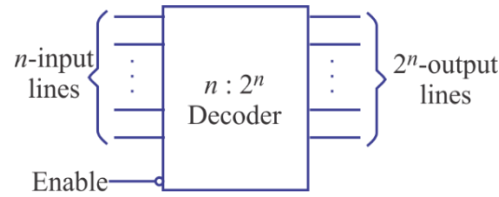


Fig. 3.26. Block Diagram of n -to- 2^n Decoder

If there are some unused or 'don't care' combinations in the n -bit code, then there will be less than 2^n output lines. In general, if n and m are respectively the numbers of input and output lines, then $m \leq 2^n$.

3.12.1. 2 to 4 Line Decoder

Consider a 2 to 4-line decoder, where A and B are two inputs whereas Y_0 through Y_3 are the four outputs.

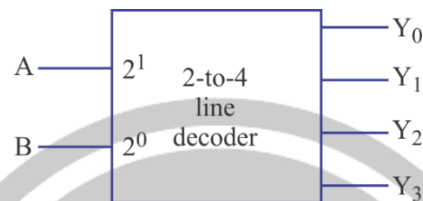


Fig. 3.27. Block Diagram of a 2 to 4 Line Decoder

Truth Table of a 2 to 4 Line Decoder

Table : Truth Table of a 2 to 4 Line Decoder

Inputs		Outputs			
A	B	Y_0	Y_1	Y_2	Y_3
0	0	1	0	0	0
0	1	0	1	0	0
1	0	0	0	1	0
1	1	0	0	0	1

The Boolean expressions for the four outputs is given as:

$$Y_0 = \bar{A}\bar{B} \text{ and } Y_1 = \bar{A}B$$

$$Y_2 = A\bar{B} \text{ and } Y_3 = AB$$

Realization of a 2 to 4 Line Decoder using Logic Gates

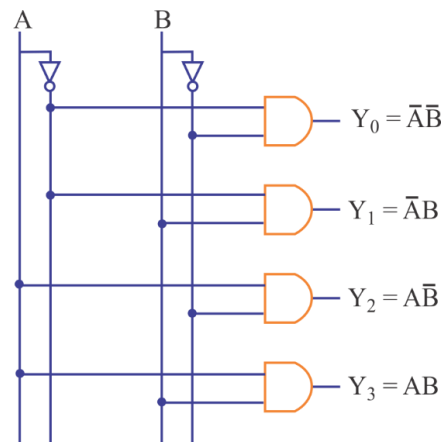


Fig. 3.28. Logic Diagram of a 2 to 4 Line Decoder

3.12.2. Applications of Decoder

Some of important applications of decoder are as follows:

1. When the decoder inputs come from a counter which is being continually pulsed, the decoder outputs will be activated sequentially. Hence, they can be used as timing or sequencing signals to turn devices on or off at specific times.
2. Decoder are use in memory system of a computer where they respond to the address code generated by the microprocessor to activate a particular memory location.
3. They are also used in computers for selection of external devices that include printers, modems, scanners, internal disk drives, keyboard, video monitor etc.

3.13. Encoders

An encoder is a combinational logic circuit that performs the inverse operation of a decoder. An encoder has 2^n (or fewer) input lines and n output lines.

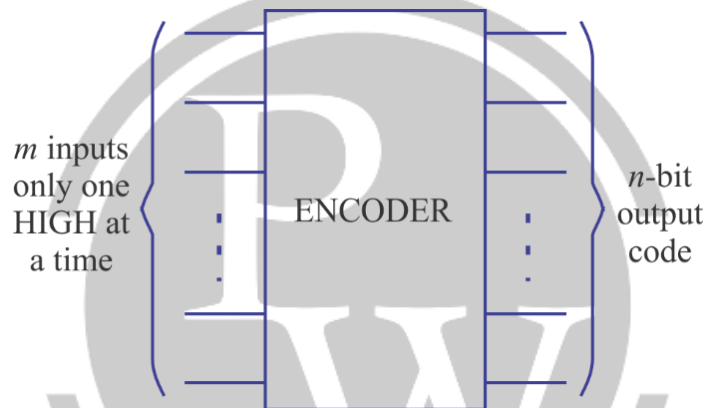


Fig. 3.29. Block Diagram of Encoder

3.13.1. Octal to Binary Encoder

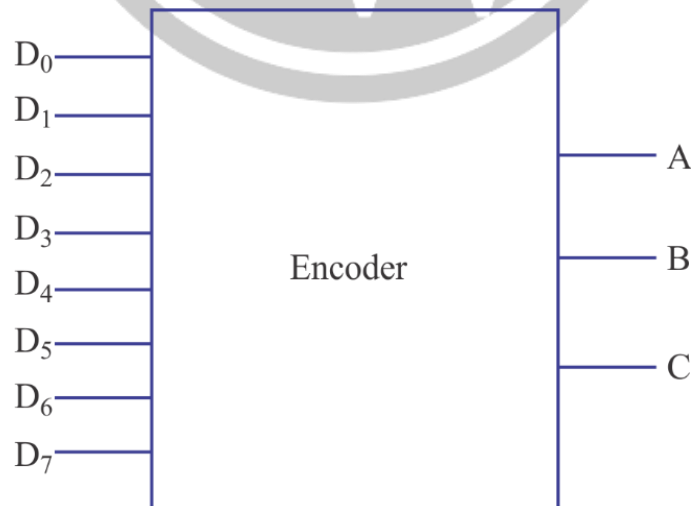


Fig. 3.30. Octal to Binary Encoder

Truth Table of an Octal to Binary Encoder:

Table: Truth Table of an Octal to Binary Encoder

Inputs								Outputs		
D ₀	D ₁	D ₂	D ₃	D ₄	D ₅	D ₆	D ₇	A	B	C
1	0	0	0	0	0	0	0	0	0	0
0	1	0	0	0	0	0	0	0	0	1
0	0	1	0	0	0	0	0	0	1	0
0	0	0	1	0	0	0	0	0	1	1
0	0	0	0	1	0	0	0	1	0	0
0	0	0	0	0	1	0	0	1	0	1
0	0	0	0	0	0	1	0	1	1	0
0	0	0	0	0	0	0	1	1	1	1

The logical expressions for the outputs as follows:

$$A = D_4 + D_5 + D_6 + D_7$$

$$B = D_2 + D_3 + D_6 + D_7$$

$$C = D_1 + D_3 + D_5 + D_7$$

3.13.2. Octal to Binary Encoder

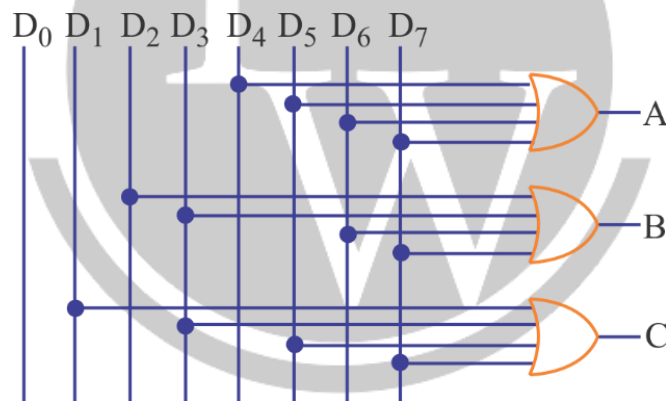


Fig. 3.31. Logic Diagram of Octal-to Binary Encoder

3.13.3. Decimal to BCD Encoder

This type of encoder has 10 inputs one for each decimal digit and 4 outputs corresponding to the BCD code.

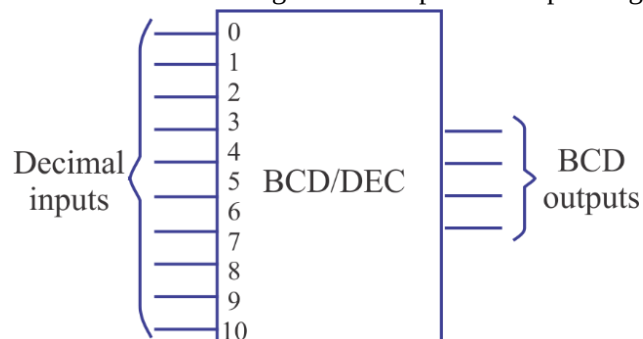


Fig. 3.32. Block Diagram of a Decimal-to-BCD Encoder

Truth Table of a Decimal to Binary Encoder:

Table : Truth Table of a Decimal to Binary Encoder

Input										Output			
0	1	2	3	4	5	6	7	8	9	A	B	C	D
D ₀	D ₁	D ₂	D ₃	D ₄	D ₅	D ₆	D ₇	D ₈	D ₉				
1	0	0	0	0	0	0	0	0	0	0	0	0	0
0	1	0	0	0	0	0	0	0	0	0	0	0	1
0	0	1	0	0	0	0	0	0	0	0	0	1	0
0	0	0	1	0	0	0	0	0	0	0	0	1	1
0	0	0	0	1	0	0	0	0	0	0	1	0	0
0	0	0	0	0	1	0	0	0	0	0	1	0	1
0	0	0	0	0	0	1	0	0	0	0	1	1	0
0	0	0	0	0	0	0	1	0	0	0	1	1	1
0	0	0	0	0	0	0	0	1	0	1	0	0	0
0	0	0	0	0	0	0	0	0	1	1	0	0	1

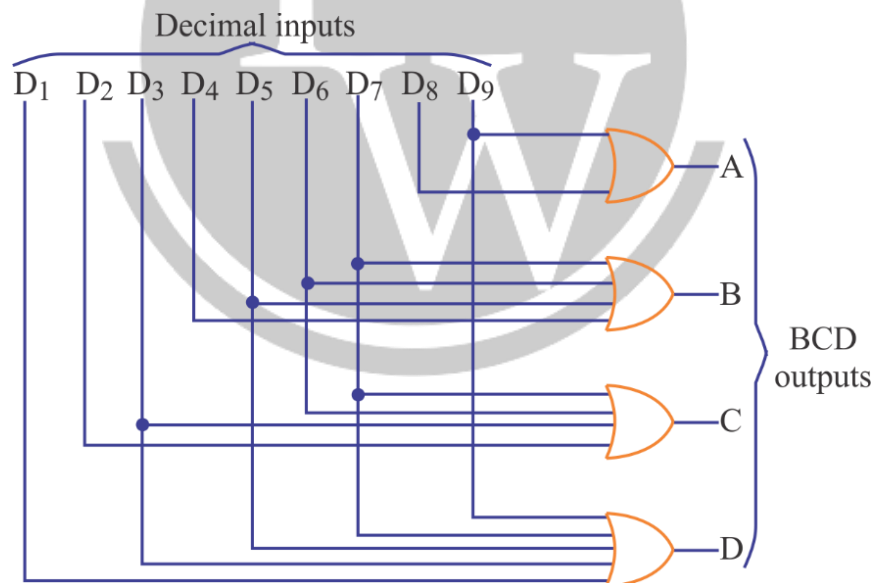


Fig. 3.33. Logic Diagram of Decimal-to-BCD encoder

The outputs of a decimal-to-BCD encoder:

$$A = D_8 + D_9$$

$$B = D_4 + D_5 + D_6 + D_7$$

$$C = D_2 + D_3 + D_6 + D_7$$

$$D = D_1 + D_3 + D_5 + D_7 + D_9$$

3.14. Priority Encoder

3.14.1. Truth Table of a Four Input Priority Encoder: (Taking LSB as priority)

Table: Truth Table of a Four Input Priority Encoder

Inputs				Outputs	
D ₀	D ₁	D ₂	D ₃	A	B
0	0	0	0	X	X
1	0	0	0	0	0
X	1	0	0	0	1
X	X	1	0	1	0
X	X	X	1	1	1

According to the truth table, the higher the subscript number, the higher the priority of the input.

The X's are don't care conditions indicating that the binary values they represent may be equal to 0 or 1.

3.15. Code Converters

A code converter is a combinational logic circuit which accepts the input information in one binary code, converts it and produces an output into another binary code.

3.15.1. The truth table for 4-bit Binary and its Equivalent BCD

Table: Truth table for 4-bit Binary and its Equivalent BCD

Decimal	Binary Input				BCD Output				
	A	B	C	D	B ₄	B ₃	B ₂	B ₁	B ₀
0	0	0	0	0	0	0	0	0	0
1	0	0	0	1	0	0	0	0	1
2	0	0	1	0	0	0	0	1	0
3	0	0	1	1	0	0	0	1	1
4	0	1	0	0	0	0	1	0	0
5	0	1	0	1	0	0	1	0	1
6	0	1	1	0	0	0	1	1	0
7	0	1	1	1	0	0	1	1	1
8	1	0	0	0	0	1	0	0	0
9	1	0	0	1	0	1	0	0	1
10	1	0	1	0	1	0	0	0	0
11	1	0	1	1	1	0	0	0	1
12	1	1	0	0	1	0	0	1	0
13	1	1	0	1	1	0	0	1	1
14	1	1	1	0	1	0	1	0	0
15	1	1	1	1	1	0	1	0	1

The minimized expression of outputs are as follows:

$$B_4 = AB + AC$$

$$B_1 = \bar{A}C + ABC$$

$$B_2 = \bar{A}B + BC$$

$$B_3 = \bar{A}B\bar{C}$$

$$B_0 = D$$

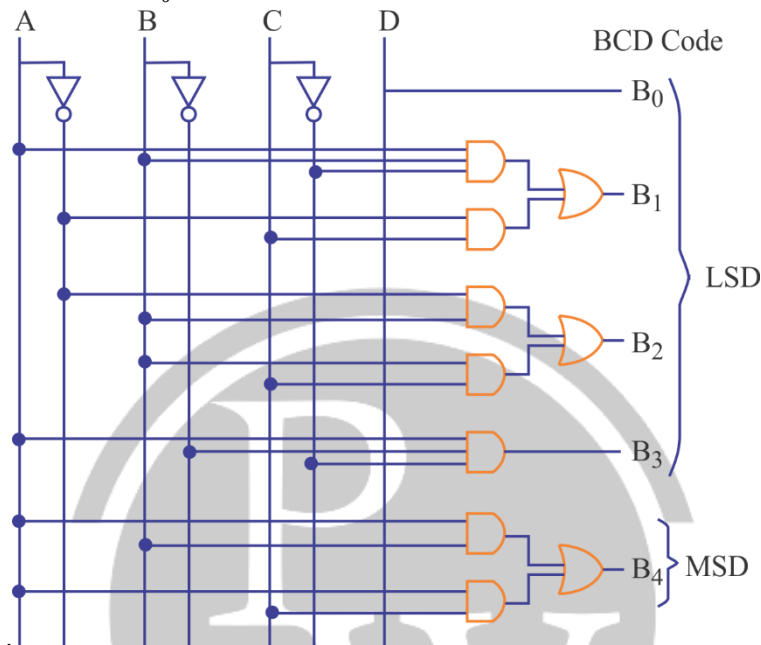


Fig. 3.34. Logic Diagram of a Binary-to-BCD Code Converter

3.16. Parity Generator

Parity generators are circuits that accept an (n-1) bit data stream and generate an extra bit that is transmitted with the bit stream. This extra bit is referred to as the parity bit. The parity added in binary message is such that the total number of 1's in the message can be either odd or even according to the type of parity used.

3.16.1. Even Parity Generator

The even parity generator is a combinational logic circuit that generates the parity bit such that the number of 1's in the message becomes even. The parity bit is '1' if there are odd number of 1's in the data stream and the parity bit is '0' if there are even number of 1's in the data stream.

Truth table for 4-bit data with Even Parity:

Table: Truth table for 4-bit data with Even Parity

4-bit data				Even Parity
A	B	C	D	P
0	0	0	0	0
0	0	0	1	1
0	0	1	0	1
0	0	1	1	0

0	1	0	0	1
0	1	0	1	0
0	1	1	0	0
0	1	1	1	1
1	0	0	0	1
1	0	0	1	0
1	0	1	0	0
1	0	1	1	1
1	1	0	0	0
1	1	0	1	1
1	1	1	0	1
1	1	1	1	0

The minimized expression for even parity generator is

$$P = A \oplus B \oplus C \oplus D$$

The logic diagram for the even parity generator is given as

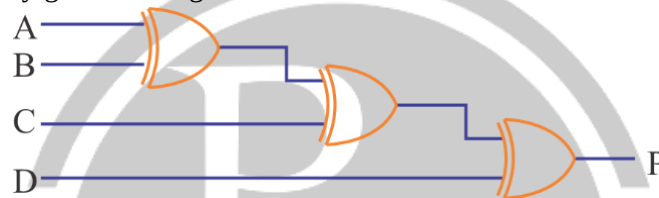


Fig. 3.35. Logic diagram of even parity generator

3.16.2. Odd Parity Generator

The odd parity generator is a combinational logic that generates the parity bit such that the number of 1's in the message becomes odd. The parity bit is '0' for odd number of 1's and '1' for even number of 1's in the bit stream.

Example: A parity generation circuit required to generate an odd parity bit may use ———?

Solution: Odd parity generation circuit consists of combination of EX-OR and EX-NOR gates, whereas even parity generator consists only EX-OR gates.

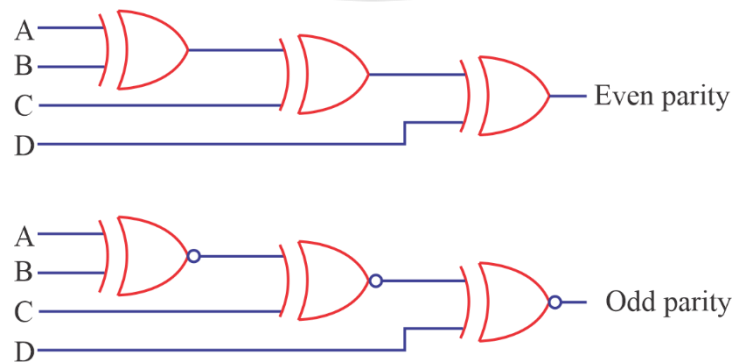


Fig. 3.36

∴

It is combination of EX-OR and EX-NOR gates.

